

3. Reading and Summarising Research Papers

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Lecture Outline

- What's in a research paper?
- Reading a research paper.
- Summarising a research paper.
- Organising your papers, notes and summaries.
- Plagiarism.

Research Papers

You **write** papers to share your findings.

- What you did, why you did it, how you did it, what you found.

You **read** papers to understand the state-of-the-art.

- They tell you what others have learned and how they did it.

Not all papers are *good* papers;

- So **critical analysis** is an important skill to develop.

Research Papers

Scientific papers typically have similar structure:

- **Abstract:** a brief synopsis of the contents;
- Introduction: what it is about, motivation, significance of the work;
- Background: prior work, context for research, background information;
- Methods: describe research artefacts and experimental design;
- Results: analysis of experiment results;
- Discussion & conclusions: interpreting the results and their implications.

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Research Papers

Lifecycle of a research paper:

- Start of a new research project;
- Perform work and write about it (in sequence or at same time);
- Revise and finish the paper;
- Submit for publication (conference, journal, book, etc)
- Peer review
- If accepted for publication:
 - Appears in conference proceedings, journal, book, etc
 - Others read it and hopefully build on it
- If not accepted for publication:
 - Revise and start again

Paper Contributions

Type	Description
Survey	Surveys and unifies a body of knowledge
Application	Reports on the development of a new system
Breakthrough	Solves a long-standing problem or presents a new way of thinking
Ground-breaking	Lays the foundations for a new field or improves our understanding of a certain area
Progress	Raises and solves new problems in an established field
Reprise	Revisits existing research with a new perspective, for new insight
Thinking	Extends a known result with a novel methodology
Debugging	Identifying and addressing flaws in existing research

Source: Ian Parberry: *A Guide for New Referees in Theoretical Computer Science*. Information and Computation 112(1), 1994.

Recap on Research Papers

Research papers make a broad variety of contributions, but all serve the same purpose: to present **methods** and **findings**.

Papers typically have a consistent **structure**;

- By following this, the right information appears in an expected place;
- This also makes it easier to **read papers methodically**.

Reading Research Papers

Reading a research paper is not always straightforward;

- Unfamiliar words, unfamiliar concepts, unfamiliar writing conventions;
- Papers are usually targeted at the authors' peers;
 - So they assume a basic understanding and shared knowledge.
- Unintuitively, you won't always read a paper in sequence;
 - And you'll probably also read it more than once.

This lecture presents one (of many) strategies for reading a paper;

- See <https://www.sciencemag.org/careers/2016/03/how-seriously-read-scientific-paper> for more ideas about how to approach research papers.

Reading Research Papers

1. Read the **title** and **abstract**;

- What is the paper about?
- Is it relevant to you?
- A good abstract should give a brief synopsis of the methods and findings, so should help you decide if it is worth reading fully.

2. Read the **introduction**;

- Why is the research important?
- What is the rest of the paper about?
- A good introduction should motivate the research and explain why the contribution is important.

Reading Research Papers

3. Read the **conclusion**;

- What conclusions are drawn from the findings?
- What contribution does this paper make?
- Jumping right to the end gives you an idea of what to expect when you go back and read the rest of the paper.

4. Skim through the rest of the paper;

- Pay attention to section headings and the structure of the paper;
- This helps build an understanding of the paper's contents;
 - Examples: what areas of related work are covered?; how many experiments were there?; does the paper present any new systems/algorithms/etc?

Reading Research Papers

5. Look at **interesting figures or charts**;
 - What do these tell us about the paper's contribution?
 - Examples: do they illustrate how a new input device works?; do they describe a new algorithm?; what are the key results?
6. *[If an experimental paper]* Read the **methodology** section;
 - What did they do?
7. Now go back to the start and read the paper in sequence;
 - **Skip the difficult bits!** Don't understand? Move on, go back to it later.
 - Highlight key points, add notes (things you don't understand, questions, etc), identify things you might need to follow up on.

Reading Research Papers

8. Now go back and **work through the difficult bits**;
 - Hopefully they make more sense now!
9. Go through the paper in full
 - You've already worked through it now, identified the important parts and worked through the difficult bits.

Summarising Papers

Actively engage with a paper as you read it:

- Highlight, annotate, take notes, sketch, work through examples, etc.
- This gives greater insight and helps you understand the content.

Reading papers actively takes time;

- You don't have time to repeat if you need to go back to a paper in future;
- There are 600+ papers in my Mendeley library, my summaries from a few years ago are still helpful today.
- So writing a good **paper summary** is important!
 - Do it after you've finished reading every paper.

Summarising Papers

There are many types of paper summary:

Type	Title	Description
Level 1	Annotated Bibliography	A brief summary of a paper, its relevance to your research, and your opinion of it.
Level 2	Summary	A longer and more detailed summary of the paper.
Level 3	Critique	A critical analysis of the paper.
Level 4	Peer Review	Similar to Level 3, but with a recommendation to an editor.
Level 5	Literature Survey	Covers several papers, reflection with an integrated narrative.

Source: Helen Purchase's five levels of paper summaries.

Level 1: Annotated Bibliography

Type	Title	Description
Level 1	Annotated Bibliography	A brief summary of a paper, its relevance to your research, and your opinion of it.

- **Meta-data**: title, authors, source, year, DOI, etc;
- **Summary paragraph**: a few sentences describing what the paper is about, its methods and key findings and conclusions;
- **Relevance**: a couple of sentences about why you read the paper, and why it is/isn't relevant to your research;
- **Personal view**: a few sentences about whether you think this is a good/poor paper (and why), what you learned, if it may be important to your research, what you'll do next.

Level 1: Annotated Bibliography

Create an entry for **every paper you read**, even if it doesn't seem completely relevant or even good;

- Because it might turn out to be relevant later;
- And you've already done the difficult bit (reading it).

Your annotated bibliography can be created within reference management software or as a separate document or database;

- More on reference management software later;
- Ideally something you can easily search through.

Level 2: Summary

Type	Title	Description
Level 2	Summary	A longer and more detailed summary of the paper.

If a paper is relevant or important to your research, produce a longer and **more detailed** summary.

- Writing this summary **consolidates your understanding** of its contents;
- You'll refer back to this in future to help understand methods, findings, implications, etc.

Level 2: Summary

Summary: brief synopsis of the paper.

Context: why did you read it and what have you learned?

Problem: what problem does the paper address?

Contribution: what are its claimed contributions?

Related work: briefly describe direct relevance to other papers you've read, or identify new papers to follow up on?

Methods: what methods were used to evaluate the solution?

Outcomes: what are the key findings of the paper?

Conclusions: what are the implications of this work – for the field and for your own research?

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Level 2: Summary

This summary template is fairly comprehensive...

- But as you read more papers, you'll know what is most relevant to you;
- Drop less relevant notes, add new ones.

Don't overestimate your memory...

- Detailed and structured summaries might seem excessive at this earlier stage, but you'll thank yourself later on.
- After weeks, months, years, you'll have read a lot of papers!

Use your summaries to inspire your writing;

- What do the best and most important papers in your field have in common?

Level 3: Critique

Type	Title	Description
Level 3	Critique	A critical analysis of the paper.

This is a **critical analysis** of a paper;

- Reflect on the **importance** of the paper's contributions;
- Evaluate the **appropriateness** of methods and **validity** of the results;
- Assess the **quality** of the paper.

As before, begin with a **short summary** of the paper;

- Short Level 1 summaries are always helpful.

Level 3: Critique

What **problem** does the paper address?

- Is it an **important** problem?
- Is this a novel solution or an incremental improvement?
- What is the **significance** of the results and conclusions?

What are the claimed **novel** contributions of the paper?

- Do you agree with the claims?

What **previous work** is this research based on?

- Does the paper show good awareness of past research?
- Is it clear what is **novel** here compared to existing work?
- Does the approach here **contradict** existing approaches?

Level 3: Critique

What **methods** have been used?

- Are they **appropriate**?
- Has the analysis been performed **correctly**?
- Is the analysis **valid**?
- Note: more on experimental design and analysis in later lectures.

What **conclusions** are drawn from the results?

- Are they **correct**?
- Does the evidence **support** the claims?

Level 4: Peer Review

Type	Title	Description
Level 4	Peer Review	Similar to Level 3, but with a recommendation to an editor.

Peer review is a key part of the scientific publication process.

- Quality, relevance and significance judged by experts in your area.
- As an academic, it is a professional obligation to (**voluntarily**) participate in the peer review process.

Outline of peer review process:

- Submit your paper to a journal or conference;
- Journal editor or conference program chair assigns expert reviewers;
- Reviewers make a recommendation to the editor/committee

Level 4: Peer Review

Peer review is basically an **extended critical analysis**;

- Each paper is critically evaluated (Level 3);
- But is also subjected to additional criteria.

Is the paper **appropriate** for this venue?

- Would conference attendees, or journal readers, be interested?
- Is the **quality** appropriate?
 - Research is often published at different levels of maturity:
 - Early research / breaking findings (workshops, conference posters)
 - A complete piece of work (full conference papers)
 - Substantial and polished paper (journal articles)
 - Note: depends on field; e.g., the top HCI venue is the annual *CHI* conference

Level 4: Peer Review

Are the correct conclusions drawn from the results?

- Think about Lecture 2 on inference...
- Does the data support the conclusions?
- Have the authors missed anything?

Is the paper presented in a satisfactory way?

- Quality of research is key, but **writing quality** also matters.
- Is the structure and style appropriate?
- Is it easy to understand the paper?
- In short: **is this ready to be published** for a wider audience to read it?

Level 4: Peer Review

Is the research **timely**?

- Is this something journal readers or conference attendees will presently be interested in?

How do readers **benefit** from this paper?

- Have the authors synthesised their findings to create recommendations or highlight implications for other research?
- What can the reader learn from this paper?

Level 4: Peer Review

Critical analysis is important when critically reviewing a paper.

Don't just critique the paper, **suggest improvements!**

- Be constructive: suggest how to address your concerns.

But, different venues have different priorities:

- E.g., a workshop paper can be expected to have a smaller contribution;
- E.g., a poster paper might describe work in progress without full results;
- E.g., a conference paper might describe a single experiment whereas a journal article contains a series of experiments.

Peer Review Considerations

Peer review is generally **voluntary**, expected of everyone who participates in a research community.

- Your peers need to review your papers, so you review theirs too.
 - Example: 3000 papers submitted to CHI 2019 so 12,000 reviews were needed!
- Peer review is balanced against other professional responsibilities;
 - Therefore not all reviews are as **thorough** as would be ideal.
- Reviewers typically learn as they go, no formal review training;
 - Leads to **inconsistency** and variance in review quality.
 - A good program committee member or editor should ensure fairness and quality, but this isn't always the case.

Peer Review Considerations

As a *reviewer*, you should:

- Be **objective**;
 - Don't let your own bias affect your judgement.
- Be **fair**;
 - Behave like you would expect your own reviewers to.
- Be **professional**;
 - Authors might not know your identity, but the editors do!
- Maintain **confidentiality**;
 - Research is submitted for peer review in confidence, don't share it.
- Be **honest**;
- Show **courtesy** to the authors;
 - Be constructive, suggest how to improve the paper, back up your criticism.

Peer Review Considerations

As an *author*, you should:

- Attribute the work of others appropriately;
 - i.e., **do not plagiarise** or take credit for others ideas.
- Be fair when critiquing the work of others;
 - It's professional courtesy, but they might also review your paper!
- Only submit to one venue at a time;
 - Simultaneous submission is bad practice, typically not allowed.
- Only publish a piece of work once;
 - However, allowances are typically made for conference papers to be revised into a more substantial journal article (at a higher standard).

Level 5: Literature Survey

Type	Title	Description
Level 5	Literature Survey	Covers several papers, reflection with an integrated narrative.

A literature survey tells a ‘**story**’ about a body of research;

- Synthesised by critically summarising several papers;
- Not just a list of papers!
 - Instead, a meaningful narrative that describes, compares, evaluates, analyses.

Lecture 4 is about searching for papers and writing literature surveys/reviews, so more on this next time...

Recap on Paper Summaries

Paper summaries play different roles in the research process.

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Source: Helen Purchase's five levels of paper summaries.

Organising Notes & Reviews

Find a way of organising notes and summaries that works for you.

Reference managers are **sortable** and **searchable** paper libraries;

- Simplify bibliography generation (e.g., using **BibTeX**)
- Typically support notes, pdf annotation, etc

Popular reference managers:

- Mendeley: <https://www.mendeley.com/>
- Zotero: <https://www.zotero.org/>
- EndNote: <https://endnote.com/>

Research Journals

Organising your paper summaries and notes is important, because it makes it easier to find relevant information in future.

On a similar note, keep a **research journal** too;

- Develop your **ideas**, keep notes on the **bigger picture**, synthesise **summaries** of the papers you're reading, describe **problems** you encounter and your **solutions** to them, etc.
- Writing helps to **organise your thoughts** – even if you don't look back at it often, the act of writing can be helpful!

Hints for Reading Papers

First, decide if a paper is **relevant**:

- Look at **title** and **context** (e.g., journal or conference name);
- Read **abstract** to see if it looks relevant, check **introduction** to make sure.

Next, get a feel for the paper:

- Skim through it, skip complex or technical content, focus on figures and results.
- Is this paper worth more of your time?

Read it fully, taking notes as you go:

- Annotate, highlight, take notes whilst you read;
- If you don't understand something, skip it and come back later!

Adapted from <https://www.csee.umbc.edu/~mariedj/691b/paper-summary-guidelines.html>

Hints for Reading Papers

After you've finished reading, update your notes:

- Write your summary of the paper and update your journal.

If a paper is **highly relevant**, **important** in your research area, or if you need to **replicate** its contribution, methods, or analysis:

- Be extra thorough in your notes and write a Level 2 or 3 summary;
- Read through the paper again, focusing on fully understanding it;
- Look at the papers it cites and **read them if they're relevant**.

Adapted from <https://www.csee.umbc.edu/~mariedj/691b/paper-summary-guidelines.html>

Plagiarism

*“the practice of taking someone else's work or ideas and **passing them off as one's own**”* (Oxford English Dictionary)

Plagiarism includes:

- Direct quotation others' work (i.e., copy and pasting);
- Paraphrasing (i.e., changing the wording but keeping the core ideas);
- Failing to attribute sources properly.

See: <https://www.gla.ac.uk/myglasgow/leads/students/plagiarism/>

Plagiarism

When you read papers, write reviews, update your notes and update your research journal:

- Properly **acknowledge** and **identify** the source for ideas;
- Properly **identify** and **attribute** direct quotes;
 - Place in quotation marks “...” and cite the source.

Reviewers, journal editors, etc., are experts in their research area:

- There's a good chance they'll review the work that plagiarises them!

See ACM Plagiarism Policy for example of professional consequences:

<https://www.acm.org/publications/policies/plagiarism-overview>

Lecture Summary

Research papers communicate the methods and findings of a piece of scientific work.

Summarise papers as you read them;

- Be **critical**: valid methods, correct conclusions, significant findings, etc?;
- Be **thorough**: effort now saves time later;
- Be **reflective**: why did you read it, what did you learn?;
- Be **organised**: reference managers help you sort and search.

Lecture Summary

There are different levels of paper summary;

- **Always write a Level 1 summary** (annotated bibliography);
- For most papers, write a Level 2 summary (detailed outline of paper);
- For important papers, write a Level 3 summary (critical analysis);

Critical review is an important research skill to develop;

- Reflect on the **importance** of the work, the **appropriateness** of its methods, the **validity** of its results, and **significance** of its contributions.
- **Peer review** is a crucial part of the scientific publication process.

Lecture Summary

Read papers methodically and review them consistently.

- Always **summarise** a paper in your own words;
 - A brief synopsis helps in future by giving a quick overview of the content.
- Write a **structured review** on the content;
 - E.g., its claimed contributions, methods, conclusions.
 - Be reflective: why did you read it, what did you learn, what would you do next?
- **Organise** your notes;
 - Reference managers, research journals, etc.

Be wary of **plagiarism**.

- If you plagiarise others in your notes, you might unintentionally use that plagiarised content in your papers too, with serious consequences.